Dataset Analysis for Hybrid Indoor Localization

WiFi + Magnetic Sensor Fingerprinting

SURA Research Project

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The Indoor Localization Challenge

Global Positioning System (GPS) technology, while revolutionary for outdoor navigation, faces significant limitations when deployed indoors. The satellite signals required for GPS triangulation are severely attenuated by building materials such as concrete, steel, and glass, resulting in unreliable or completely unavailable positioning information within indoor environments.

This fundamental limitation has driven extensive research into alternative localization methodologies that leverage signals of opportunity already present in indoor spaces. The emergence of ubiquitous wireless infrastructure and sophisticated smartphone sensors has created new possibilities for accurate indoor positioning systems.

WiFi Received Signal Strength

Radio signal measurements from wireless access points throughout the building

Magnetic Field Disturbances

Unique magnetic signatures created by indoor metal structures and building materials

Inertial Measurement Unit (IMU)

Accelerometer, gyroscope, and orientation sensors tracking device movement

These complementary sensing modalities can be fused together to create robust indoor localization systems that overcome the individual limitations of each signal type. The integration of multiple sensor streams provides redundancy, improves accuracy, and enables continuous positioning even when individual signal sources become temporarily unavailable.

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Research Objective and Dataset

Project Goal

Develop a high-accuracy indoor localization system leveraging hybrid sensor data fusion from WiFi, magnetic field, and inertial sensors

This research employs the **MagWi Benchmark Dataset**, a comprehensive collection of smartphone sensor data gathered from indoor environments across multiple buildings. The dataset represents one of the most extensive collections of its kind, specifically designed for benchmarking indoor localization algorithms and sensor fusion techniques.

The dataset was collected using multiple smartphone models operated by different users, ensuring diversity in hardware characteristics and usage patterns. This multi-user, multi-device approach enhances the generalizability of any localization models developed from this data.

- ### WiFi Received Signal Strength (RSS)

  Signal measurements from detected wireless access points with timestamps and coordinates

- ### Magnetometer Readings

  Three-axis magnetic field measurements capturing spatial variations in magnetic signatures

- ### Accelerometer Data

  Three-axis acceleration measurements for motion detection and activity recognition

- ### Gyroscope Measurements

  Angular velocity data for orientation tracking and movement characterization

- ### Pressure Sensor

  Atmospheric pressure readings for floor-level detection in multi-story buildings

- ### Orientation Sensors

  Device orientation and heading information derived from sensor fusion algorithms

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Dataset Composition and Structure

Data File Overview

The dataset comprises two primary categories of sensor recordings, each serving a distinct purpose in the localization framework. The organization follows a timestamp-based structure where each file represents sensor readings at a specific point in time and physical location.

Temporal Characteristics

The sampling frequency differs significantly between WiFi scanning and inertial/magnetic sensors, reflecting their distinct physical properties and power consumption characteristics.

2831

WiFi Scan Files

Individual fingerprint locations with associated access point detections

4261

Magnetic/IMU Recordings

High-frequency sensor measurements synchronized with WiFi scans

50–200

IMU Sampling Rate (Hz)

Continuous high-frequency measurements of inertial and magnetic signals

1–2

WiFi Scan Rate (Hz)

Periodic wireless network discovery operations balancing accuracy with power consumption

The disparity in sampling rates between WiFi and IMU sensors is intentional and reflects practical constraints. WiFi scanning consumes significant power and generates radio frequency emissions, limiting how frequently smartphones can perform network discovery operations. Conversely, inertial and magnetic sensors can sample continuously with minimal power overhead, providing rich temporal information about device movement and environmental characteristics.

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WiFi Data Format and Organization

Column Name	Description
Time	Timestamp of the WiFi scan measurement in Unix epoch format with millisecond precision
X-pos	Cartesian X-coordinate representing the physical location where the scan was taken (in meters)
Y-pos	Cartesian Y-coordinate representing the physical location where the scan was taken (in meters)
SSID	Service Set Identifier - the human-readable name of the wireless network broadcast by the access point
BSSID	Basic Service Set Identifier - the unique MAC address (hardware identifier) of the wireless router or access point
RSS	Received Signal Strength - power level of the WiFi signal measured in decibels relative to one milliwatt (dBm)

Each WiFi fingerprint file follows a structured format that combines location metadata with wireless network observations. The file structure employs a hierarchical organization where the first row establishes the spatial reference point, and subsequent rows enumerate detected access points at that location.

This file organization has important implications for data preprocessing and feature engineering. When constructing training datasets for machine learning models, all rows from a single file must be aggregated to create a comprehensive fingerprint representation for that location. The aggregation process typically involves creating a feature vector where each element represents the signal strength from a specific access point, with missing values handled through imputation or zero-padding strategies.

Critical Data Structure Insight

Only the first row in each file contains valid X and Y coordinate values. All subsequent rows represent additional access points detected during the same scan at the identical physical location. Therefore, each file corresponds to exactly one fingerprint location, but may contain multiple access point detections.

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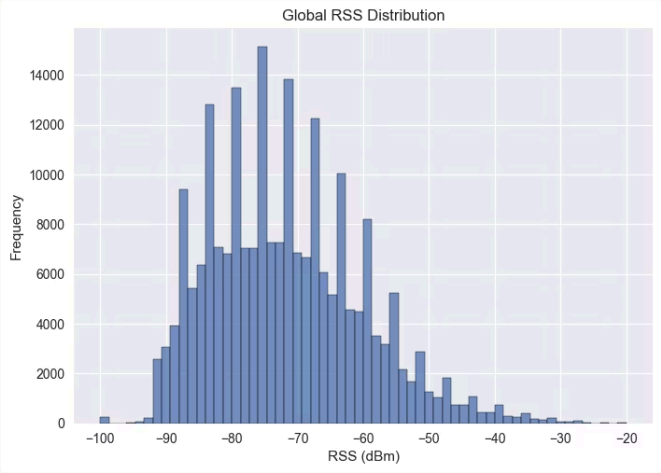
Received Signal Strength (RSS) Distribution Analysis

-72	-95	-20	11.5
Mean RSS (dBm)	Minimum RSS (dBm)	Maximum RSS (dBm)	Standard Deviation



The histogram of RSS values reveals a distribution centered around -72 dBm with a standard deviation of approximately 11.5 dBm, indicating substantial signal variability across the measurement locations. The minimum observed value of -95 dBm represents the practical detection threshold where signals become indistinguishable from background noise, while the maximum of -20 dBm indicates proximity to powerful access points with direct line-of-sight propagation paths.

The signal distribution falls within expected indoor ranges, validating the dataset quality. Very few missing values (<0.1%) suggest reliable WiFi scanning hardware and stable wireless infrastructure during data collection.

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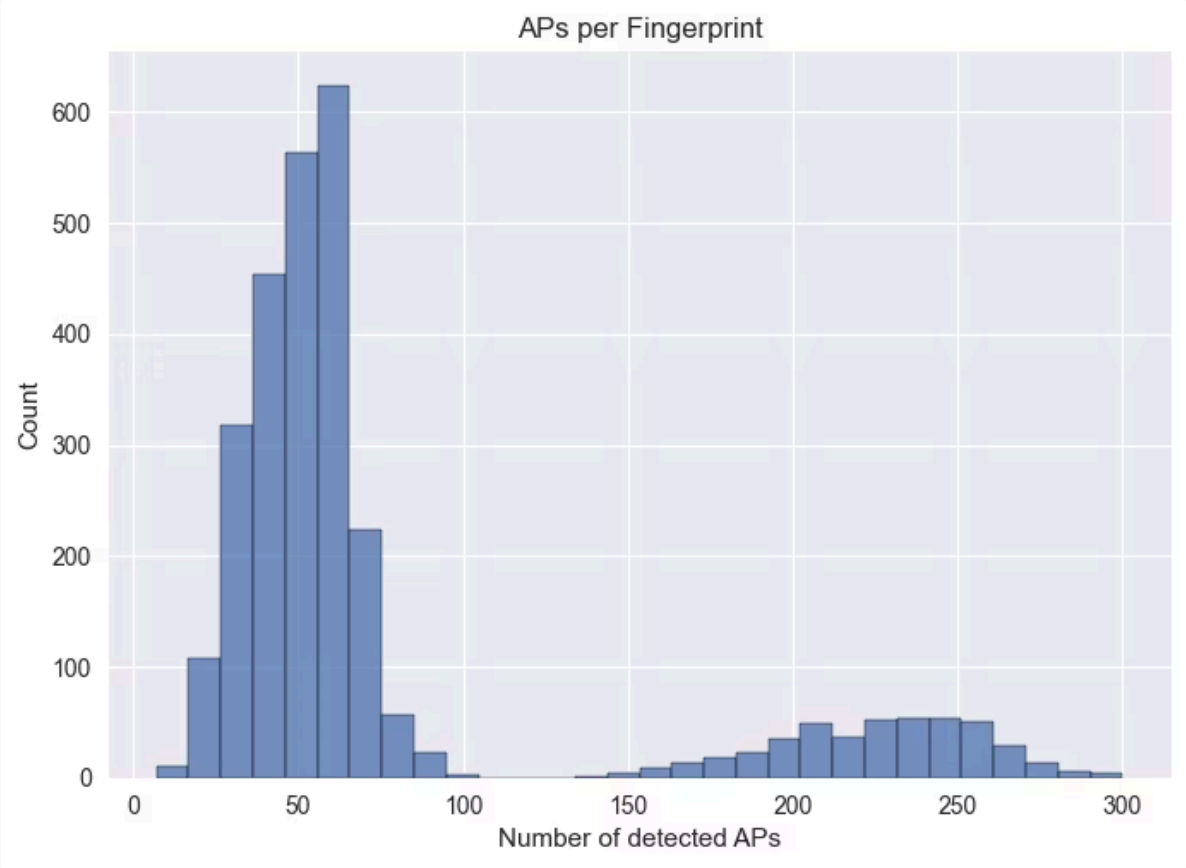
Access Point Density and Distribution



78

Average APs per Scan

300

Maximum APs

7

Minimum APs

Access point density varies significantly (7-300 APs per location) due to building layout and infrastructure differences. Higher density improves localization accuracy by providing more spatial features, but excessive density adds computational overhead. Of 1425 unique routers detected, some appear 1400+ times (stable features), while many appear rarely (potential noise). Feature selection can identify discriminative access points and filter unreliable routers

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Spatial Distribution and Trajectory Patterns

Scatter plot visualization of the fingerprint locations reveals 2381 labeled fingerprints collected along a continuous trajectory through the indoor environment. The spatial distribution exhibits a distinctive pattern where the Y-coordinate remains nearly constant while the X-coordinate varies significantly, suggesting data collection along a corridor or hallway path.

Continuous Trajectory

2381 labeled fingerprint locations arranged along a path, indicating systematic data collection rather than random sampling

Corridor Path Pattern

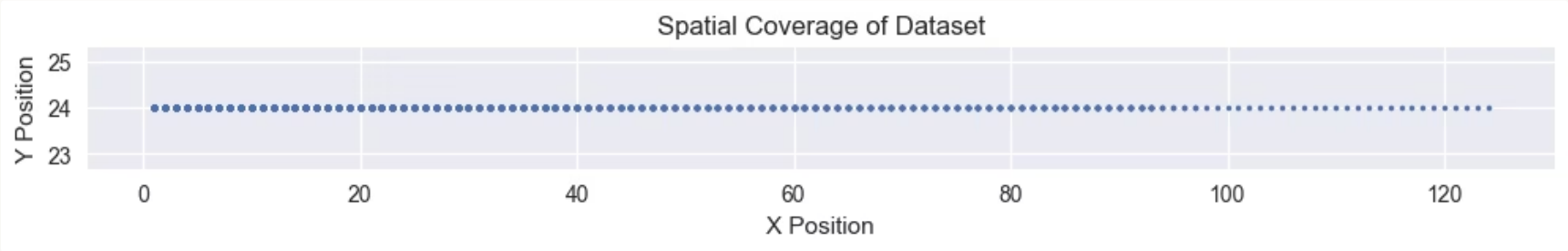
Y-coordinate remains almost constant while X varies, suggesting linear hallway or corridor collection route

Sequential Dependencies

Temporal ordering of measurements enables sequential modeling approaches using LSTM or GRU architectures

The trajectory nature of the data collection has important implications for model selection and evaluation. Traditional classification or regression models treat each fingerprint as an independent observation, ignoring the temporal dependencies between consecutive measurements. However, sequential models such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) can exploit the temporal structure, using previous locations and movements to inform current position estimates.

This sequential pattern also suggests opportunities for trajectory smoothing and dead reckoning integration, where inertial sensor data between WiFi scans can be used to interpolate positions and improve temporal resolution. The combination of fingerprint matching with inertial dead reckoning represents a common sensor fusion approach in indoor localization systems.



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Magnetic Field Characteristics

44

Mean Magnitude (µT)

7.1

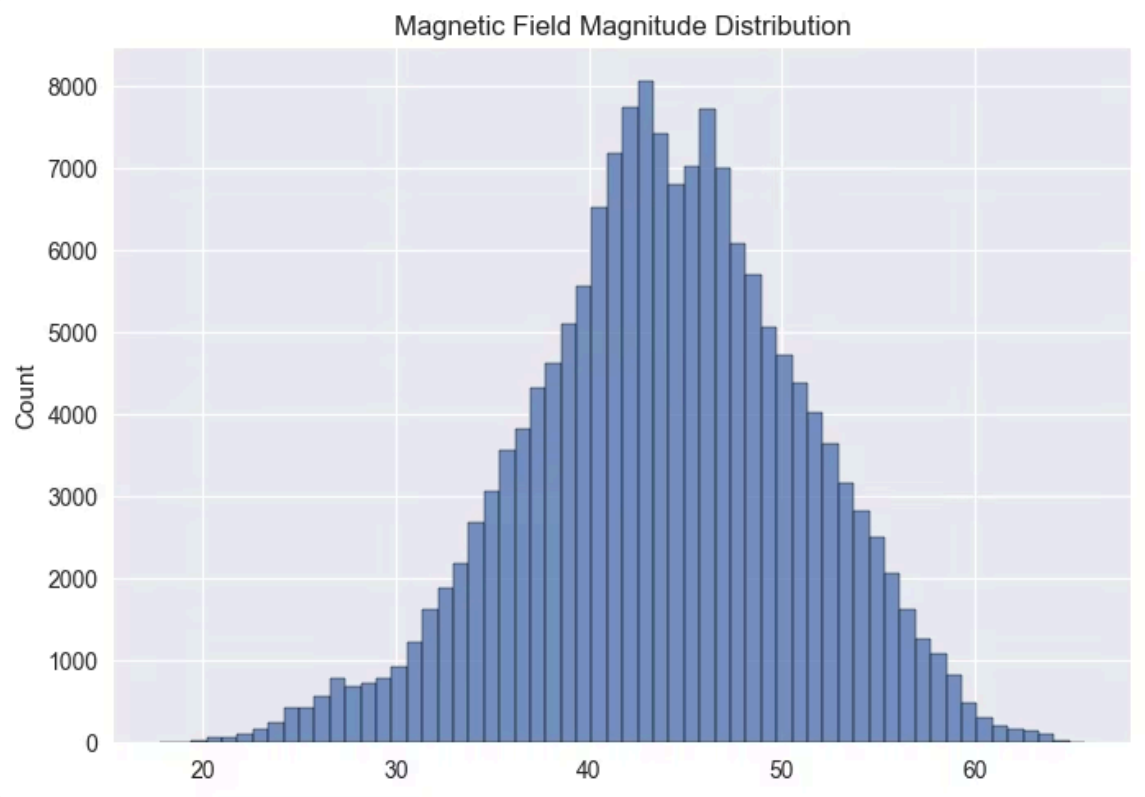
Standard Deviation (µT)

18

Minimum (µT)

66

Maximum (µT)



The magnetometer data reveals stable signal distributions with mean magnetic field magnitude of approximately 43.9 microtesla (µT) and standard deviation of 7.1 µT, spanning a range from 18 to 66 µT. These values align with typical indoor magnetic field characteristics, where the Earth's natural magnetic field (approximately 25-65 µT) is distorted by building materials, metal structures, and electrical installations.

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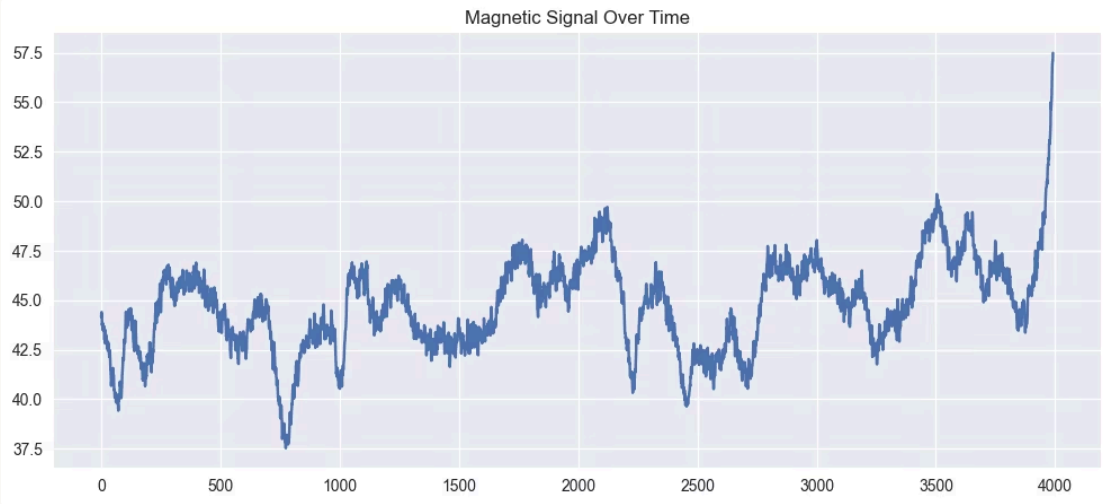
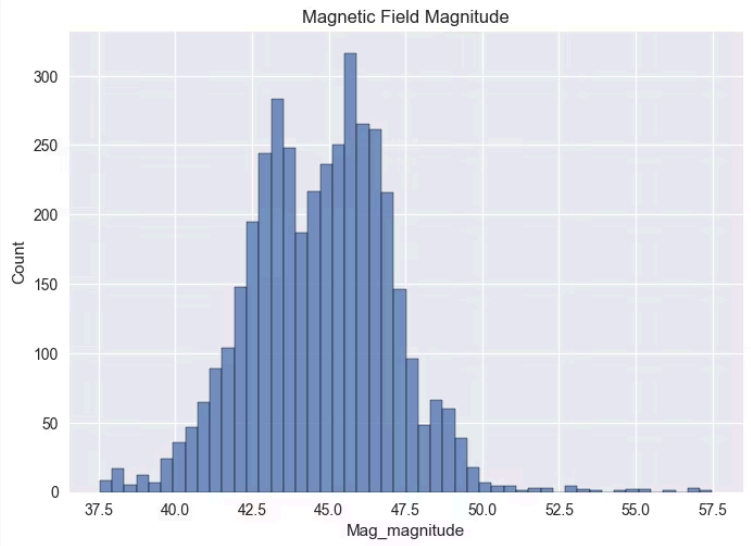
Magnetic Field Singular File



The localized magnetic distortions created by indoor metal structures serve as distinctive location fingerprints. Unlike WiFi signals that can be affected by network configuration changes, magnetic field distortions are largely determined by permanent building characteristics, making them stable over time. This temporal stability represents a significant advantage for long-term localization deployments where WiFi infrastructure may be reconfigured or replaced.



Three-axis magnetometer readings can be combined to compute magnetic field magnitude or decomposed into directional components, each providing different information for localization. The magnitude captures the overall strength of distortion at a location, while directional components reveal the spatial orientation of metal structures. Both approaches have demonstrated effectiveness in magnetic fingerprinting literature.

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Summary and Next Research Steps

Dataset Quality Assessment

The MagWi Benchmark Dataset represents a high-quality, large-scale collection suitable for developing and evaluating hybrid indoor localization systems. The dataset exhibits realistic indoor signal characteristics, high access point density, and complementary sensor modalities.

Key Findings

Statistical analysis confirms expected signal distributions, minimal missing data, and systematic spatial coverage along continuous trajectories.

WiFi Characteristics

- 2831 fingerprints with coordinates
- 1425 unique routers detected
- ~78 APs per location on average
- RSS centered at -72 dBm
- <0.1% missing values

Magnetic Signatures

- Stable signal distribution
- Mean magnitude ~44 µT
- Range 18-66 µT
- Useful for fingerprinting
- Time-invariant features

Expected Benefits of Hybrid Sensor Fusion

The combination of WiFi fingerprinting with magnetic field measurements and inertial sensors is expected to improve localization accuracy compared to single-sensor approaches. WiFi provides spatial discrimination through access point visibility patterns, magnetic fields offer time-invariant location signatures, and IMU sensors enable continuous tracking between fingerprint updates.

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Next Phase: Model Development and Evaluation

1

Feature Vector Construction

Convert raw WiFi fingerprints into structured feature vectors with consistent dimensionality, handling missing values and access point variability

2

Sensor Fusion Architecture

Design integration framework combining WiFi RSS, magnetic magnitude, and IMU-derived features into unified representation

3

Training Dataset Preparation

Construct labeled training dataset with cross-validation folds, ensuring temporal independence between training and test sets

4

Model Implementation

Implement and tune machine learning models including Random Forest, Gradient Boosting, k-Nearest Neighbors, and deep learning architectures

5

Performance Benchmarking

Compare localization accuracy against state-of-the-art methods using standard metrics and statistical significance testing

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